Conjecture by Riccardo Sparacino

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Abstract. *Let p_1 and p_2 be two consecutive primes, $d = p_2 - p_1$. Define $N = \lceil \ln^2(d \cdot p_1) \rceil$. We have experimentally verified that for all 50,847,534 pairs with $p_1 \leq 10^9$ the interval $[\, dp_1 + 1,\ dp_1 + N \,]$ always contains at least one prime number. This document presents the conjecture, explains it step-by-step with examples, and discusses its connections to the Riemann Hypothesis and Cramér's conjecture.*

General formula:

$$r = (p_2 - p_1) \cdot p_1 + k, \qquad \text{with } 1 \leq k \leq N$$

$$N = \left\lceil \ln^2\big((p_2 - p_1) \cdot p_1\big) \right\rceil$$

The search interval is

$$I = \big[\, (p_2 - p_1) \cdot p_1 + 1,\ (p_2 - p_1) \cdot p_1 + N \,\big]$$

and r is any prime number belonging to this interval.

Explanation of every symbol and step

✤ p_1 and p_2

They are two **consecutive prime numbers**, with $p_1 < p_2$. Example: $p_1 = 5$, $p_2 = 7$ (there are no primes between 5 and 7).

✤ $d = p_2 - p_1$

It is the **gap** between the two consecutive primes. In the example: $d = 7 - 5 = 2$.

✤ $M = d \cdot p_1$

Product of the gap and the smaller prime. In the example: $M = 2 \cdot 5 = 10$. This number is the starting point to which we will add a small increment.

✤ $N = \lceil \ln^2 M \rceil$

We compute the natural logarithm of M ($\ln M$), square it, and take the **ceiling**. In the example: $\ln(10) \approx 2.302585$, $\ln^2(10) \approx 5.3019$, so $N = \lceil 5.3019 \rceil = 6$. This N is the *width* of the interval where we will look for a prime.

✤ **Interval** $I = [M + 1,\ M + N]$

Starting from $M + 1$ up to $M + N$ (inclusive). In the example:
$I = [10 + 1,\ 10 + 6] = [11, 16]$.

✤ $r = M + k$

It is a prime number contained in I. The parameter k is therefore the distance from M to the found prime, and satisfies $1 \leq k \leq N$. In the example: the interval [11,16] contains the primes 11 and 13. We can choose $r = 11$ (then $k = 11 - 10 = 1$) or $r = 13$ ($k = 3$).

✤ **Primality test**

> To ensure that r is indeed prime, one can use a primality test (e.g., trial division by primes up to $\sqrt{r}$ or a probabilistic test). In the following examples we will verify manually.

The conjecture (2026 version)

Conjecture. For every pair of consecutive primes (p_1, p_2) (with $p_1 \geq 2$), the interval $I(p_1, p_2)$ contains at least one prime number. Equivalently, there exists an integer k with $1 \leq k \leq N$ such that

$$r = d\,p_1 + k$$

is prime.

Complete examples

Three general examples

📌 **Example 1 – pair (5, 7)**

Step	Operation	Result
1. p_1, p_2	consecutive primes	$p_1 = 5,\ p_2 = 7$
2. $d = p_2 - p_1$	$7 - 5$	$d = 2$
3. $M = d \cdot p_1$	2×5	$M = 10$
4. $\ln M$	natural log of 10	$\ln 10 \approx 2.302585093$
5. $\ln^2 M$	$(2.302585093)^2$	≈ 5.301898
6. $N = \lceil \ln^2 M \rceil$	ceiling	$N = 6$
7. Interval I	$[M+1, M+N] = [11, 16]$	$11, 12, 13, 14, 15, 16$
8. Primes in I	manual check	11 and 13

9. Choice of r and k	$r = 11 \Rightarrow k = 11 - 10 = 1$ or $r = 13 \Rightarrow k = 3$	$k = 1$ (minimum)
10. Verification of r	11 not divisible by 2,3,5,7 ($\sqrt{11} \approx 3.3$)	✅ 11 is prime

📌 Example 2 – pair (23, 29)

Step	Operation	Result
1. p_1, p_2	consecutive primes	$p_1 = 23,\ p_2 = 29$
2. $d = p_2 - p_1$	$29 - 23$	$d = 6$
3. $M = d \cdot p_1$	6×23	$M = 138$
4. $\ln M$	$\ln 138$	≈ 4.927253685
5. $\ln^2 M$	$(4.927253685)^2$	≈ 24.27786
6. $N = \lceil \ln^2 M \rceil$	ceil(24.27786)	$N = 25$
7. Interval I	$[139, 163]$	$139, 140, \ldots, 163$
8. Primes in I	sieve	$139, 149, 151, 157, 163$
9. Choose r	first prime in interval	$r = 139 \Rightarrow$ $k = 139 - 138 = 1$
10. Verify $r = 139$	not divisible by 2,3,5,7,11 ($\sqrt{139} \approx 11.8$)	✅ 139 is prime

📌 Example 3 – pair (113, 127)

Step	Operation	Result
1. p_1, p_2	consecutive primes (verified)	$p_1 = 113,\ p_2 = 127$

2. $d = p_2 - p_1$	$127 - 113$	$d = 14$
3. $M = d \cdot p_1$	14×113	$M = 1582$
4. $\ln M$	$\ln 1582$	≈ 7.366445
5. $\ln^2 M$	7.366445^2	≈ 54.264
6. $N = \lceil \ln^2 M \rceil$	ceil(54.264)	$N = 55$
7. Interval I	$[1583, 1637]$	length 55
8. Primes in I	search	$1583, 1597, 1601, 1607, 1609, 1613, 1619, 1621, 1627, 1637$
9. Choose r	first prime in interval	$r = 1583 \Rightarrow k = 1583 - 1582 = 1$
10. Verify $r = 1583$	not divisible by primes up to 37 ($\sqrt{1583} \approx 39.8$)	✅ 1583 is prime

Special case: twin primes ($d = 2$)

Twin primes are pairs of the form $(p, p+2)$. Here the gap is minimal ($d = 2$) and they provide a significant test for the conjecture. Below are three specific examples.

👫 Twin primes – Example A: (17, 19)

Step	Operation	Result
1. p_1, p_2	twin primes	$p_1 = 17,\ p_2 = 19$
2. $d = p_2 - p_1$	$19 - 17$	$d = 2$
3. $M = d \cdot p_1$	2×17	$M = 34$
4. $\ln M$	$\ln 34$	≈ 3.52636

5. $\ln^2 M$	3.52636^2	≈ 12.435
6. $N = \lceil \ln^2 M \rceil$	ceil(12.435)	$N = 13$
7. Interval I	$[35, 47]$	$35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47$
8. Primes in I		$37, 41, 43, 47$
9. Choose r	first prime in interval	$r = 37 \Rightarrow k = 37 - 34 = 3$
10. Verify $r = 37$	$\sqrt{37} \approx 6.08$, not divisible by 2,3,5	✅ 37 is prime

👫 Twin primes – Example B: (41, 43)

Step	Operation	Result
1. p_1, p_2	twin primes	$p_1 = 41,\ p_2 = 43$
2. $d = p_2 - p_1$	$43 - 41$	$d = 2$
3. $M = d \cdot p_1$	2×41	$M = 82$
4. $\ln M$	$\ln 82$	≈ 4.40672
5. $\ln^2 M$	4.40672^2	≈ 19.419
6. $N = \lceil \ln^2 M \rceil$	ceil(19.419)	$N = 20$
7. Interval I	$[83, 102]$	$83, 84, \ldots, 102$
8. Primes in I		$83, 89, 97, 101$
9. Choose r	first prime in interval	$r = 83 \Rightarrow k = 83 - 82 = 1$
10. Verify $r = 83$	$\sqrt{83} \approx 9.11$, not divisible by 2,3,5,7	✅ 83 is prime

👫 Twin primes – Example C: (107, 109)

Step	Operation	Result
1. p_1, p_2	twin primes	$p_1 = 107,\ p_2 = 109$
2. $d = p_2 - p_1$	$109 - 107$	$d = 2$
3. $M = d \cdot p_1$	2×107	$M = 214$
4. $\ln M$	$\ln 214$	≈ 5.365976
5. $\ln^2 M$	5.365976^2	≈ 28.793
6. $N = \lceil \ln^2 M \rceil$	ceil(28.793)	$N = 29$
7. Interval I	$[215, 243]$	$215, 216, \ldots, 243$
8. Primes in I	check	$223, 227, 229, 233, 239, 241$
9. Choose r	first prime in interval	$r = 223 \Rightarrow$ $k = 223 - 214 = 9$
10. Verify $r = 223$	$\sqrt{223} \approx 14.93$, not divisible by 2,3,5,7,11,13	✅ 223 is prime

Note: In this case the first available prime in the interval is 223 (with $k = 9$), but the interval contains as many as 6 primes.

Global numerical evidence

A C++/Python program was implemented to:

- generate all primes up to $10^9 + 2000$ using an optimized Eratosthenes sieve;
- for every consecutive pair (p_1, p_2) with $p_1 \leq 10^9$ compute $M = d\,p_1$ and $N = \lceil \ln^2 M \rceil$;
- search for a prime in the interval $[M + 1, M + N]$ using a deterministic Miller–Rabin test for 64-bit integers.

Experimental verification up to $p_1 \leq 10^9$

Range of p_1	Consecutive pairs	Successes	Failures
$2 - 10^6$	78,498	78,498	0

Range of p_1	Consecutive pairs	Successes	Failures
$2 - 10^9$	50,847,534	50,847,534	**0**

✓ **50,847,534 consecutive pairs** with $p_1 \leq 1,000,000,000$:
no counterexample – 100% of the intervals contain at least one prime.

Among these, thousands are twin prime pairs, and in every case the interval I contained at least one prime number.

Theoretical connections

Riemann Hypothesis (RH)

Under the Riemann Hypothesis it is known that there is a prime in $[x, x + c\sqrt{x}\ln x]$ for some constant $c > 0$. Our N is of order $\ln^2 x$, much smaller than $\sqrt{x}\ln x$. Therefore, if the conjecture were true, it would show that the distribution of primes is even more regular than guaranteed by RH. Conversely, a counterexample would provide an interval of length $\ln^2 M$ with no primes, an event hardly compatible with RH.

Cramér's conjecture

Cramér's conjecture states that $\limsup_{n\to\infty} \frac{p_{n+1} - p_n}{\ln^2 p_n} = 1$. In our construction, if d is large (close to $\ln^2 p_1$), the interval I widens accordingly, making it easier to find a prime. The proposed conjecture can be viewed as a “soft” version of Cramér's idea, applied to the product dp_1.

Future work

- **Extend the verification** to $p_1 \leq 10^{10}$ or 10^{11} (segmented sieve, 128-bit integers).
- **Analyze the distribution of** k (the “step” inside the interval).
- **Study correlations** with the parity of d or other arithmetic properties.
- **Attempt a conditional proof** assuming RH or other standard conjectures.

Final remarks

- The formula only requires knowledge of two consecutive primes.
- The computation of N is deterministic and depends solely on p_1 and d.
- In tests on 50.8 million pairs, the interval always contained at least one prime (100% success).
- The value of k varies: sometimes $k = 1$ (the first number in the interval is already prime), sometimes larger.
- The choice of k is not unique; there are generally several primes inside the same interval.
- For twin primes, the interval starts at $2p_1 + 1$, which is odd and often prime, but not always (e.g., (17,19) gives $k = 3$).

References

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2. Cramér, H. (1936). *On the order of magnitude of the difference between consecutive prime numbers*. Acta Arithmetica, 2, 23-46.
3. Ribenboim, P. (2004). *The Little Book of Bigger Primes*. Springer.

Note: logarithm calculations were performed with 6-digit approximations. Using the ceiling ensures the interval is always at least the computed width, even with small rounding errors.

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